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ORIGINAL REPORT

MONARCH 2: Abemaciclib in Combination With Fulvestrant in Women With HR+/HER2− Advanced Breast Cancer Who Had Progressed While Receiving Endocrine Therapy

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ABSTRACT

Purpose
MONARCH 2 (ClinicalTrials.gov identifier: NCT02107703) compared the efficacy and safety of abemaciclib, a selective cyclin-dependent kinase 4 and 6 inhibitor, plus fulvestrant with fulvestrant alone in patients with advanced breast cancer (ABC).

Patients and Methods
MONARCH 2 was a global, double-blind, phase III study of women with hormone receptor-positive and human epidermal growth factor receptor 2-negative ABC who had progressed while receiving neoadjuvant or adjuvant endocrine therapy (ET), ≤ 12 months from the end of adjuvant ET, or while receiving first-line ET for metastatic disease. Patients were randomly assigned 2:1 to receive abemaciclib or placebo (150 mg twice daily) on a continuous schedule and fulvestrant (500 mg, per label). The primary end point was investigator-assessed progression-free survival (PFS), and key secondary end points included overall survival, objective response rate (ORR), duration of response, clinical benefit rate, quality of life, and safety.

Results
Between August 2014 and December 2015, 669 patients were randomly assigned to receive abemaciclib plus fulvestrant (n = 446) or placebo plus fulvestrant (n = 223). Abemaciclib plus fulvestrant significantly extended PFS versus fulvestrant alone (median, 16.4 *v* 9.3 months; hazard ratio, 0.553; 95% CI, 0.449 to 0.681; $P < .001$). In patients with measurable disease, abemaciclib plus fulvestrant achieved an ORR of 48.1% (95% CI, 42.6% to 53.6%) compared with 21.3% (95% CI, 15.1% to 27.6%) in the control arm. The most common adverse events in the abemaciclib versus placebo arms were diarrhea (86.4% *v* 24.7%), neutropenia (46.0% *v* 4.0%), nausea (45.1% *v* 22.9%), and fatigue (39.9% *v* 26.9%).

Conclusions
Abemaciclib at 150 mg twice daily plus fulvestrant was effective, significantly improving PFS and ORR and demonstrating a tolerable safety profile in women with hormone receptor-positive and human epidermal growth factor receptor 2-negative ABC who progressed while receiving ET.

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INTRODUCTION

More than 70% of patients with metastatic breast cancer (mBC) present with hormone receptor-positive (HR+) disease and are candidates for endocrine therapy (ET), with benefit diminishing as resistance develops.[1-3] To improve the outcomes of patients whose disease progressed while receiving ET, it is necessary to understand and overcome mechanisms of ET resistance. Cyclin-dependent kinases 4 and 6 (CDK 4 and 6) inhibition is proving to be an effective strategy.[4-6]

The molecular mechanisms governing ET resistance and oncogenic growth converge at the cell cycle. Preclinical data have revealed that estrogen receptor (ER)-induced proliferation requires cyclin D,[7,8] which is highly expressed in > 50% of patients with breast cancer.[9] Cyclin D is the catalyst for CDK 4 and 6, and overactivation of CDK 4 and 6 may attenuate senescence and promote cell cycle progression.[10-14] Direct

inhibition of CDK 4 and 6 disrupts this pathway and diminishes breast cancer cell growth. Preclinically, short-term inhibition of CDK 4 and 6 induces a temporary G1 arrest that rebounds upon withdrawal[15]; however, continuous inhibition of CDK 4 and 6 has demonstrated sustained growth arrest with initiation of apoptosis or senescence.[16]

Abemaciclib is an orally administered, potent, and selective small-molecule inhibitor of CDK 4 and 6 administered on a twice daily continuous schedule.[4,10,15] Abemaciclib is structurally distinct from other CDK 4 and 6 inhibitors (such as ribociclib and palbociclib) and is 14 times more potent against cyclin D1/CDK 4 and cyclin D3/CDK 6 in enzymatic assays.[10,15-18] Abemaciclib prevents CDK 4 and 6 phosphorylation of the retinoblastoma tumor suppressor protein, thereby inducing G1 arrest and abrogating cell growth.[15] Preclinical evaluation demonstrated the antitumor activity of abemaciclib in an ER+/human epidermal growth factor receptor 2-negative (HER2−) breast cancer xenograft model.[10] In the phase I setting, abemaciclib demonstrated activity in HR+ mBC as monotherapy and in combination with fulvestrant.[10] These data prompted MONARCH 1, a phase II study of abemaciclib as a single agent (200 mg given twice daily on a continuous schedule) in patients with hormone refractory HR+/HER2− mBC, in which the investigator-assessed overall response rate was 19.7%, the median duration of response was 8.6 months, and the clinical benefit rate (CBR; complete response [CR] plus partial response [PR] plus stable disease ≥ 6 months) was 42.4%.[4]

Here, we report the results of MONARCH 2 (ClinicalTrials.gov identifier: NCT02107703), a phase III trial comparing the safety and efficacy of abemaciclib plus fulvestrant versus placebo plus fulvestrant in women with HR+/HER2− advanced breast cancer (ABC; ie, inoperable locally advanced or mBC) who progressed while receiving ET.

PATIENTS AND METHODS

Study Design and Patients

MONARCH 2 was a phase III, randomized, double-blind, placebo-controlled study of fulvestrant with or without abemaciclib in women with HR+/HER2− ABC whose disease had progressed while receiving prior ET. The study was conducted in 142 centers in 19 countries.

Eligible women were ≥ 18 years old with any menopausal status (pre- or perimenopausal women received a gonadotropin-releasing hormone agonist) and had an Eastern Cooperative Oncology Group performance status of 0 or 1. Disease had to be measurable by Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1[19] or nonmeasurable bone-only disease (ie, blastic, lytic, or mixed). Patients were required to have disease that progressed while receiving neoadjuvant or adjuvant ET, ≤ 12 months after adjuvant ET, or while receiving ET for ABC. Patients must not have received more than one ET or any prior chemotherapy for ABC. Exclusion criteria included prior treatment with fulvestrant, everolimus, or CDK 4 and 6 inhibitors; presence of visceral crisis; or evidence or history of CNS metastasis.

All patients provided informed consent before joining the study. Before the start of the trial, the protocol received ethical and institutional

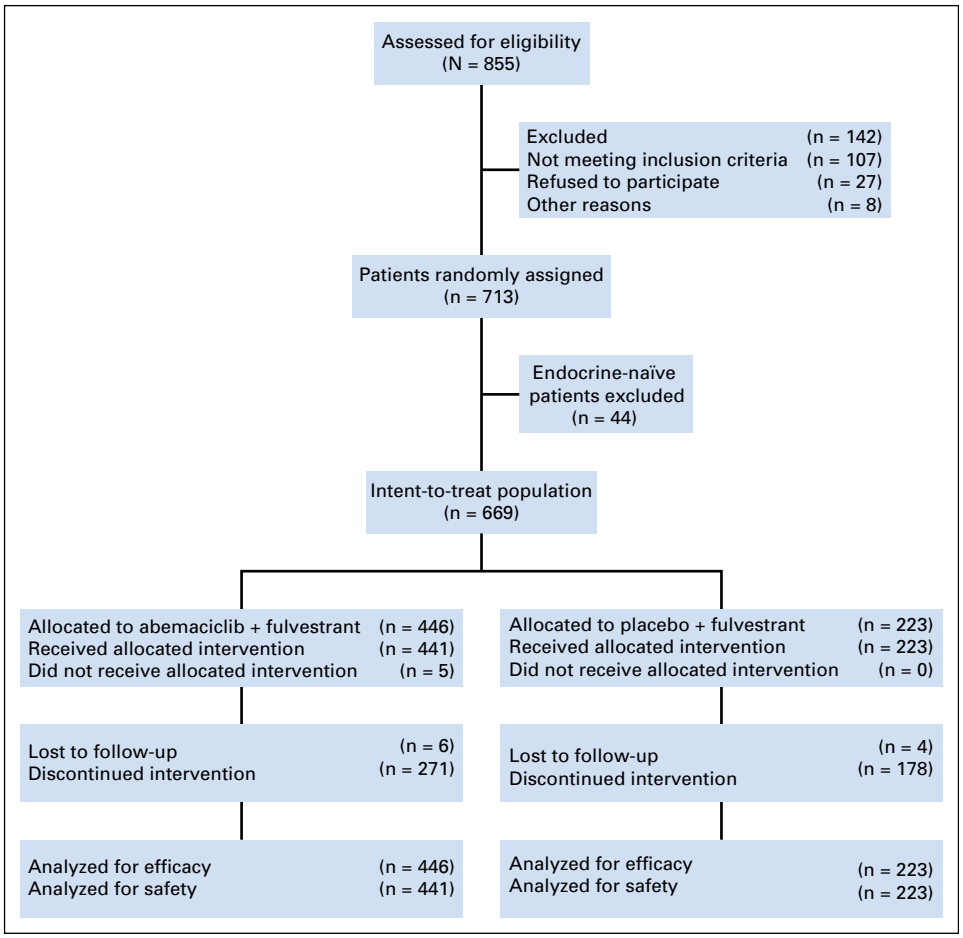

Fig 1. CONSORT diagram.

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review board approvals. This study was performed in compliance with the Declaration of Helsinki. A steering committee oversaw the conduct of the trial. An independent data monitoring committee reviewed the safety data quarterly.

Randomization and Treatment Procedures

Using an interactive, web-based randomization scheme, patients were assigned to receive abemaciclib plus fulvestrant or placebo plus fulvestrant in a 2:1 ratio. Randomized assignment was stratified according to metastatic site (visceral, bone only, or other) and ET resistance (primary or secondary). Primary ET resistance, as defined by European Society for Medical Oncology guidelines, includes patients whose disease relapsed while receiving the first 2 years of neoadjuvant or adjuvant ET or progressed while receiving the first 6 months of ET for ABC.[20,21] Patients who were not considered to have primary ET resistance were defined as having secondary resistance.

Patients received 500 mg fulvestrant by intramuscular injection on days 1 and 15 of the first cycle, and on day 1 of subsequent cycles (every 28 days). Patients received abemaciclib or placebo twice daily during each 28-day cycle. At study initiation, patients in the abemaciclib arm received 200 mg twice daily. After a review of safety data and dose reduction rates, the protocol was amended to reduce the starting dose to 150 mg for new patients, and all patients who were receiving 200 mg underwent a mandatory dose reduction to 150 mg. Treatment continued until progressive disease (PD), death, or patient withdrawal.

Dose interruptions and reductions of abemaciclib or placebo were permitted according to prespecified dose-adjustment procedures for patients who exhibited treatment-related toxicities. Fulvestrant dose reductions were permitted per US label as determined by the investigator. Patients were not permitted to switch treatment groups. If either abemaciclib or placebo was discontinued, patients were permitted to continue receiving fulvestrant; if fulvestrant required discontinuation, patients were permitted to continue receiving abemaciclib or placebo.

Efficacy and Safety Measures

Tumors were measured by computed tomography or magnetic resonance imaging according to RECIST version 1.1 within 28 days before random assignment (baseline) and then every 8 weeks the first year, every 12 weeks thereafter, and within 2 weeks of clinical progression. Patients underwent bone scintigraphy at baseline, and then again every sixth cycle starting with cycle 7. Hematologic and blood chemistry laboratory tests were performed centrally on days 1 and 15 of the first cycle and day 1 of all remaining cycles.

Table 1. Patient and Disease Baseline Characteristics

Characteristic	Abemaciclib + Fulvestrant (n = 446)	Placebo + Fulvestrant (n = 223)
Age, years, median (range)	59 (32-91)	62 (32-87)
ET resistance*		
Primary	111 (24.9)	58 (26.0)
Secondary	326 (73.1)	163 (73.1)
Most recent ET†		
Neoadjuvant or adjuvant	263 (59.0)	133 (59.6)
Metastatic	171 (38.3)	85 (38.1)
Prior AI		
Yes	316 (70.9)	149 (66.8)
No	130 (29.1)	74 (33.2)
PgR status‡		
Positive	339 (76.0)	171 (76.7)
Negative	96 (21.5)	44 (19.7)
Metastatic site§		
Visceral	245 (54.9)	128 (57.4)
Bone only	123 (27.6)	57 (25.6)
Other	75 (16.8)	38 (17.0)
Measurable disease		
Yes	318 (71.3)	164 (73.5)
No	128 (28.7)	59 (26.5)
Race‖		
Asian	149 (33.4)	65 (29.1)
Caucasian	237 (53.1)	136 (61.0)
Other	29 (6.5)	13 (5.8)
ECOG performance status¶		
0	264 (59.2)	136 (61.0)
1	176 (39.5)	87 (39.0)
Prior chemotherapy for neoadjuvant or adjuvant treatment		
Yes	267 (59.9)	134 (60.1)
No	179 (40.1)	89 (39.9)
Menopausal status#		
Pre- or perimenopause	72 (16.1)	42 (18.8)
Postmenopause	371 (83.2)	180 (80.7)

Note. Data given as No. (%) unless otherwise indicated.
Abbreviations: AI, aromatase inhibitor; ECOG, Eastern Cooperative Oncology Group; ET, endocrine therapy; PgR, progesterone receptor.
*Six patients in the abemaciclib arm and two patients in the placebo arm had not received prior ETs.
†ET history was not available for 12 patients in the abemaciclib arm and five patients in the placebo arm.
‡Eight patients in each arm had unknown PgR status.
§Metastatic site was not available for three patients in the abemaciclib arm.
‖A total of 31 patients in the abemaciclib arm and nine in the placebo arm had missing race information.
¶One patient had ECOG performance status of 2 in the abemaciclib arm.
#Menopausal status was not available for three patients in the abemaciclib arm and one in the placebo arm.

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Adverse events (AEs) were recorded and graded according to the National Cancer Institute Common Terminology Criteria, version 4.0, and were evaluated at every patient visit from baseline until follow-up. All AEs were characterized by severity and whether they were related to the study drug.

Outcomes

The primary end point, investigator-assessed PFS, was analyzed from the time of random assignment until objective PD or death for any reason. Key secondary end points included objective response rate (ORR; ie, proportion of patients with CR or PR), duration of response (time from CR or PR until PD or death), CBR, and safety and tolerability. Other secondary end points not reported in this analysis are overall survival (OS), quality of life measures, and pharmacokinetics.

Statistical Analysis

The study was designed to compare the PFS for abemaciclib plus fulvestrant to that for placebo plus fulvestrant. The study initially planned to enroll 450 patients into the intent-to-treat (ITT) population. However, after a change in the starting dose of the blinded-study drug from 200 mg to 150 mg, the sample size was increased to 630 patients to ensure at least 450 patients were enrolled at the 150-mg dose. The primary statistical analyses for investigator-assessed PFS were performed on the ITT population, which included all patients regardless of starting dose. Sensitivity analyses were planned that (1) included only patients enrolled after the change in starting dose and that (2) determined progression on the basis of a blinded, independent central review.

The primary end point, investigator-assessed PFS, was evaluated using a log-rank test stratified by metastatic site and ET resistance. The final analysis was planned at 378 PFS events, which would provide approximately 90% power assuming a hazard ratio (HR) of 0.703 at a one-sided α of 0.025, which corresponds to a 2.75-month improvement over the true median PFS for the control arm of 6.5 months.[22] One efficacy interim analysis was planned to be at 70% of the final PFS events. The α control for the secondary end point of OS was maintained using a hierarchical testing approach.

The odds ratio estimator and the stratified Cochran-Mantel-Haenszel test were used to compare the rates of binary end points. Two-sided *P* values were used to compare efficacy between treatment groups and for interaction tests associated with the subgroup factors. An exploratory mixed-model analysis was used to compare change in tumor size over time. Unless otherwise noted, all hypothesis tests were performed at the two-sided .05 level, and all confidence intervals used a 95% confidence level. Safety was assessed in all patients who received at least one dose of study drug (ie, the safety population). SAS (version 9.2 or later; SAS Institute, Cary, NC) was used for statistical analyses.

RESULTS

Patients

From August 7, 2014, to December 29, 2015, 669 patients were randomly assigned to receive abemaciclib plus fulvestrant (n = 446) or placebo plus fulvestrant (n = 223; Fig 1). The data cut off occurred February 14, 2017. Patients had well-balanced baseline characteristics (Table 1). At baseline, 373 patients (55.8%) presented with visceral disease and 180 (26.9%) with bone-only disease. A total of 169 patients (25.3%) had primary ET resistance, and 18 (2.7%) had locally advanced disease; 140 (20.9%) patients were progesterone receptor-negative. Most patients entered the study after progressing while receiving prior ET (8.8% of patients progressed within 12 months after completing adjuvant therapy).

Treatment

At the data cut off, 170 patients (38.1%) in the abemaciclib arm versus 45 (20.2%) in the placebo arm were continuing to receive the study drug. Patients in the abemaciclib arm received a median of 15 cycles compared with nine cycles in the control arm. Patients who received 200 mg of abemaciclib before the mandatory dose-reduction amendment (n = 121; 27.4%) received a median of 34 days of drug before dose reduction or discontinuation. Dose intensities were 273.1 mg/d in the abemaciclib arm compared with 298.2 mg/d in the placebo arm.

Abemaciclib was discontinued for AEs in 70 patients (15.9%) versus seven patients (3.1%) in the placebo arm. The abemaciclib dose was reduced because of AEs in 189 patients (42.9%) compared with three (1.3%) receiving placebo. Abemaciclib was interrupted

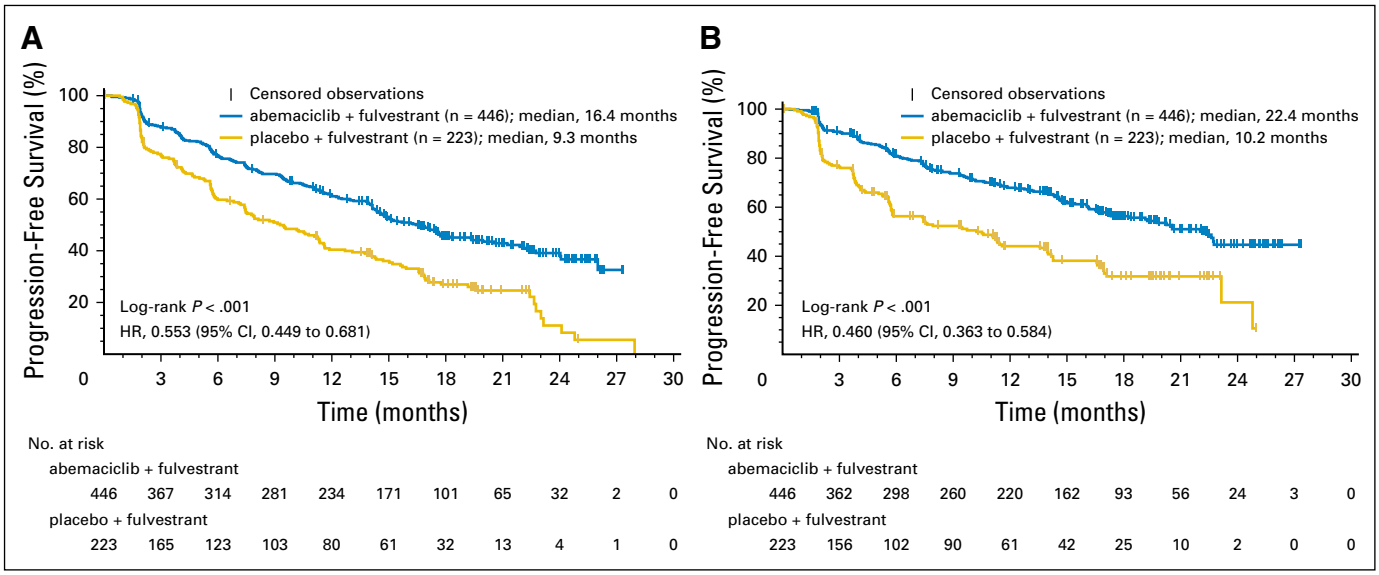

Fig 2. Kaplan-Meier plots of progression-free survival. (A) Investigator-assessed and (B) independent central review of intent-to-treat population. HR, hazard ratio.

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because of AEs in 229 patients (51.9%) and in 26 patients (11.7%) in the placebo arm.

Efficacy

In the ITT population, 379 PFS events (documented progression or death without documented progression) occurred (n = 222 [49.8%] in the abemaciclib plus fulvestrant arm and n = 157 [70.4%] in the control arm). The median length of follow-up was 19.5 months. The abemaciclib plus fulvestrant arm achieved a median PFS of 16.4 months compared with 9.3 months in the control arm (HR, 0.553; 95% CI, 0.449 to 0.681; $P < .001$; Fig 2A). A blinded central analysis demonstrated consistent PFS results

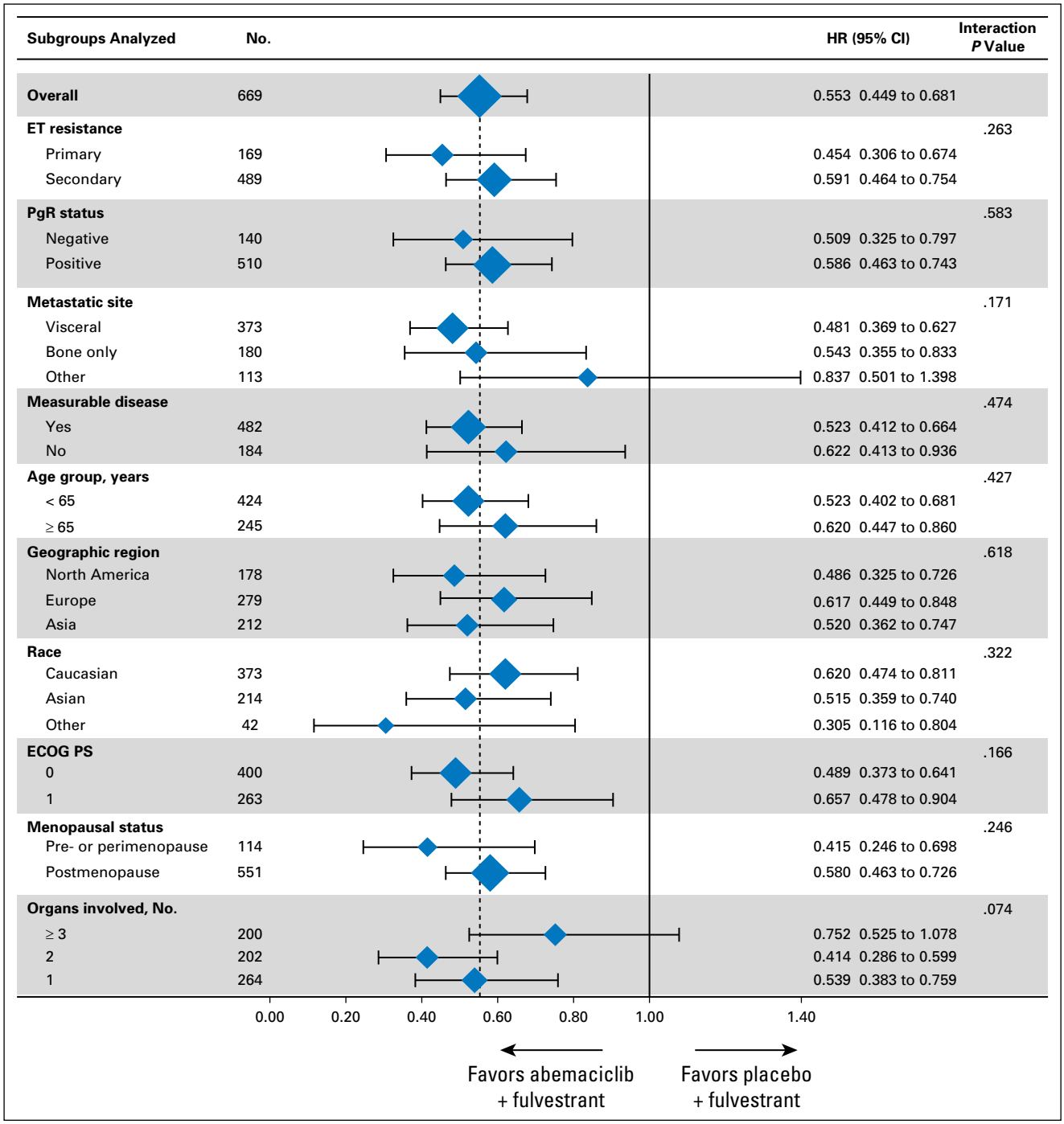

Fig 3. Progression-free survival (PFS) of patient subgroups. HRs are for abemaciclib versus placebo. *P* values are for the interaction term from a model with arm, the subgroup variable and arm × subgroup interaction term. PFS HRs are indicated by diamonds and 95% CIs are indicated by the crossing horizontal lines. Diamond size is proportional to each patient subgroup population size. HRs are unstratified and estimated with the adjustment of arm × subgroup interaction, except the overall PFS. The overall PFS estimates were stratified by metastatic site and ET resistance. The factor levels that consisted of < 5% of randomly assigned patients were omitted from the analysis. ECOG PS, Eastern Cooperative Oncology Group performance status; ET, endocrine therapy; PgR, progesterone receptor.

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(HR, 0.460; 95% CI, 0.363 to 0.584; $P < .001$; Fig 2B). A sensitivity analysis including only those patients enrolled after the change in starting dose was consistent with the ITT analysis (HR, 0.588; 95% CI, 0.458 to 0.754). The addition of abemaciclib to fulvestrant improved PFS across all patient subgroups (Fig 3).

The ORR in the ITT population was 35.2% (95% CI, 30.8% to 39.6%) in the abemaciclib arm and 16.1% (95% CI, 11.3% to 21.0%) in the control arm ($P < .001$; Table 2). This included 14 CRs (3.1%) in the abemaciclib arm compared with one CR (0.4%) in the control arm. Responses in both arms were durable, with 12-month duration of response rates of 67.8% in the abemaciclib arm and 66.9% in the placebo arm. The median duration of response had not been reached in the abemaciclib arm, with 90 responders (57.3%) continuing to receive treatment at the time of the analysis. Patients with measurable disease achieved an ORR of 48.1% (95% CI, 42.6% to 53.6%) in the abemaciclib arm and 21.3% (95% CI, 15.1% to 27.6%) in the control arm ($P < .001$).

In addition, an exploratory analysis of change in tumor size over time was conducted. After 12 cycles of treatment, the mean change in tumor size for the abemaciclib arm was −62.5%, compared with −32.8% for the placebo arm (Fig 4). The CBR was 72.2% (95% CI, 68.0% to 76.4%) in the abemaciclib arm and 56.1% (95% CI, 49.5% to 62.6%; $P < .001$) in the placebo arm. Best response of PD was lower in the abemaciclib arm than the placebo arm (9.0% *v* 20.2%). At the time of the data cut off, OS results were not yet mature, with 85 deaths (19.1%) in the abemaciclib arm and 48 (21.5%) in the placebo arm.

Safety

In the safety population (abemaciclib, n = 441; placebo, n = 223), the most frequent adverse events of any grade were diarrhea, neutropenia, nausea, fatigue, and abdominal pain (Table 3). These occurred at predominately grade 1 or 2 severity. Febrile neutropenia was reported in six patients in the abemaciclib arm. Of these cases, one patient had grade 2 afebrile neutropenia miscoded as febrile neutropenia, and one patient had febrile neutropenia 53 days after discontinuing abemaciclib and had received poststudy paclitaxel before the event. The four remaining cases of febrile neutropenia were not associated with severe infection (grade ≥ 3). Granulocyte colony-stimulating factor use was low in both arms (Appendix Table A1, online only). There was a higher incidence of infections in the abemaciclib arm (42.6%) than in the placebo arm (24.7%) regardless of relatedness; however, these infections were predominately of grade 1 to 2 severity (6.6% in the abemaciclib arm *v* 3.6% in the placebo arm were grade ≥ 3).

Serious adverse events (SAEs) were reported in 22.4% of patients in the abemaciclib arm and 10.8% of patients in the placebo arm. SAEs possibly related to the study drug were reported in 8.8% of patients on the abemaciclib arm and 1.3% of patients on the placebo arm, with the most frequent being diarrhea (1.4% in the abemaciclib arm *v* 0% in the placebo arm). Thromboembolic events were the most frequently reported SAE and occurred in nine patients (2.0%) in the abemaciclib arm and one patient (0.4%) in the placebo arm. Of the patients in the abemaciclib arm, four experienced an SAE of pulmonary embolism, none of which resulted in death.

Grade 1 or 2 diarrhea occurred in 322 patients (73.0%) in the abemaciclib arm and 54 (24.2%) in the control arm. In contrast, grade 3 diarrhea was less frequent (n = 59 [13.4%] *v* n = 1 [0.4%] in the abemaciclib and control arms, respectively). In the abemaciclib arm, diarrhea events typically occurred in the first

Table 2. Best Overall Response

Best Overall Response*	Abemaciclib + Fulvestrant (n = 446)		Placebo + Fulvestrant (n = 223)		Odds Ratio	*P* Value
	No. (%)	95% CI†	No. (%)	95% CI†		
Intent-to-treat population						
CR	14 (3.1)	1.5 to 4.8	1 (0.4)	−0.4 to 1.3		
PR	143 (32.1)	27.7 to 36.4	35 (15.7)	10.9 to 20.5		
SD	213 (47.8)	43.1 to 52.4	133 (59.6)	53.2 to 66.1		
≥ 6 months	165 (37.0)	32.5 to 41.5	89 (39.9)	33.5 to 46.3		
Progressive disease	40 (9.0)	6.3 to 11.6	45 (20.2)	14.9 to 25.4		
Not evaluable	36 (8.1)	5.5 to 10.6	9 (4.0)	1.5 to 6.6		
Overall response rate (CR + PR)	157 (35.2)	30.8 to 39.6	36 (16.1)	11.3 to 21.0	2.82	< .001
Disease control rate (CR + PR + SD)	370 (83.0)	79.5 to 86.4	169 (75.8)	70.2 to 81.4	1.56	0.025
Clinical benefit rate (CR + PR + SD ≥ 6 months)	322 (72.2)	68.0 to 76.4	125 (56.1)	49.5 to 62.6	2.04	< .001
Measurable disease						
CR	11 (3.5)	1.5 to 5.5	0	NA		
PR	142 (44.7)	39.2 to 50.1	35 (21.3)	15.1 to 27.6		
SD	109 (34.3)	29.1 to 39.5	84 (51.2)	43.6 to 58.9		
≥ 6 months	80 (25.2)	20.4 to 29.9	50 (30.5)	23.4 to 37.5		
Progressive disease	32 (10.1)	6.8 to 13.4	38 (23.2)	16.7 to 29.6		
Not evaluable	24 (7.5)	4.6 to 10.5	7 (4.3)	1.2 to 7.4		
Overall response rate (CR + PR)	153 (48.1)	42.6 to 53.6	35 (21.3)	15.1 to 27.6	3.42	< .001
Disease control rate (CR + PR + SD)	262 (82.4)	78.2 to 86.6	119 (72.6)	65.7 to 79.4	1.77	0.012
Clinical benefit rate (CR + PR + SD ≥ 6 months)	233 (73.3)	68.4 to 78.1	85 (51.8)	44.2 to 59.5	2.55	< .001

Abbreviations: CR, complete response; PR, partial response; SD, stable disease.
*Using RECIST version 1.1.
†CIs were based on normal approximation.

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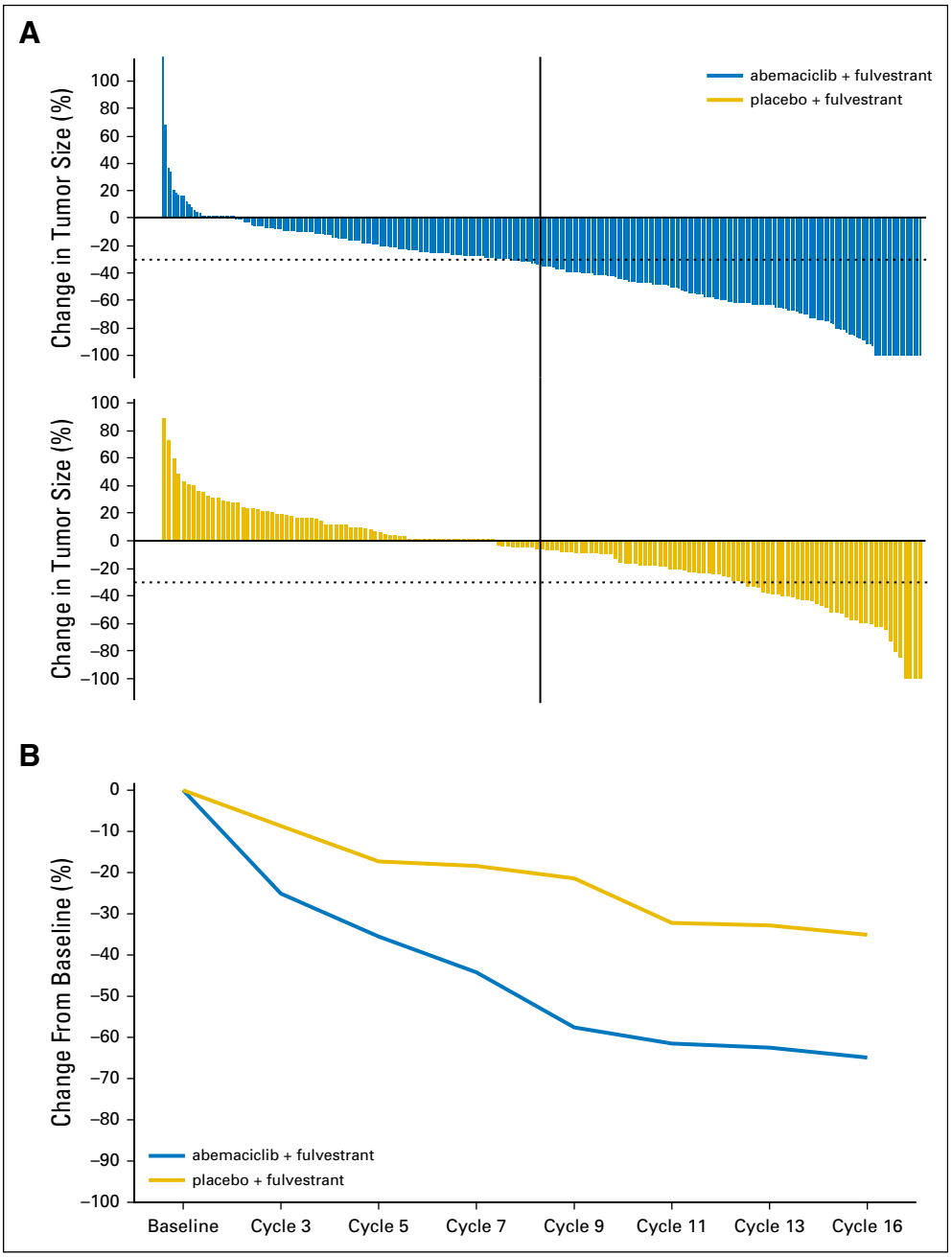

Fig 4. Tumor response. (A) Best percent change in tumor size for all evaluable patients. Dotted horizontal line (−30%) corresponds to the partial response criteria defined by RECIST version 1.1. The vertical line is a visual to locate approximately 50% of patients. (B) Percentage change in tumor size from baseline by cycle (at day 1 of each cycle).

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treatment cycle (median onset of diarrhea was 6 days). In most cases, diarrhea was effectively managed using antidiarrheal medications and with dose adjustments. In the abemaciclib arm, 14.5% of patients who experienced an initial grade 2 diarrhea event and 1.1% of patients who experienced an initial grade 3 diarrhea event also experienced a recurrence at the same or higher grade. The majority (70.1%) of patients in the abemaciclib arm with an AE of diarrhea did not require treatment modification (ie, dose interruption, reduction, or discontinuation); however, 2.9% of patients discontinued study drug because of diarrhea.

On the basis of central laboratory analysis, the most common abnormalities were increased serum creatinine level, decreased WBC and neutrophil counts, and anemia. Approximately 25% more patients in the abemaciclib arm experienced an increase in serum creatinine level than those in the placebo arm (Appendix Table A2, online only). Abemaciclib has been shown to increase serum creatinine levels due to inhibition of renal tubular secretion of creatinine without affecting glomerular function.[23]

There were 14 deaths (3.2%) in the abemaciclib arm (nine due to AEs) and 10 (4.5%) in the control arm (two due to AEs) while those patients were receiving therapy or within 30 days of treatment discontinuation. Of these, three deaths (0.7%) in the abemaciclib arm were determined to be related to the study treatment; two were due to sepsis in patients in whom guidance

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Table 3. Treatment-Emergent Adverse Events

	CTCAE Grade, No. (%)					
	Abemaciclib + Fulvestrant (n = 441)			Placebo + Fulvestrant (n = 223)		
Adverse Event (≥ 10% in either arm)	All	3	4	All	3	4
Any	435 (98.6)	241 (54.6)	26 (5.9)	199 (89.2)	46 (20.6)	5 (2.2)
Diarrhea	381 (86.4)	59 (13.4)	0	55 (24.7)	1 (0.4)	0
Neutropenia	203 (46.0)	104 (23.6)	13 (2.9)	9 (4.0)	3 (1.3)	1 (0.4)
Nausea	199 (45.1)	12 (2.7)	—	51 (22.9)	2 (0.9)	—
Fatigue	176 (39.9)	12 (2.7)	—	60 (26.9)	1 (0.4)	—
Abdominal pain	156 (35.4)	11 (2.5)	—	35 (15.7)	2 (0.9)	—
Anemia	128 (29.0)	31 (7.0)	1 (0.2)	8 (3.6)	2 (0.9)	0
Leukopenia	125 (28.3)	38 (8.6)	1 (0.2)	4 (1.8)	0	0
Decreased appetite	117 (26.5)	5 (1.1)	0	27 (12.1)	1 (0.4)	0
Vomiting	114 (25.9)	4 (0.9)	0	23 (10.3)	4 (1.8)	0
Headache	89 (20.2)	3 (0.7)	—	34 (15.2)	1 (0.4)	—
Dysgeusia	79 (17.9)	—	—	6 (2.7)	—	—
Alopecia	69 (15.6)	—	—	4 (1.8)	—	—
Thrombocytopenia	69 (15.6)	9 (2.0)	6 (1.4)	6 (2.7)	0	1 (0.4)
Stomatitis	67 (15.2)	2 (0.5)	0	23 (10.3)	0	0
Constipation	60 (13.6)	3 (0.7)	0	30 (13.5)	1 (0.4)	0
ALT increased	59 (13.4)	17 (3.9)	1 (0.2)	12 (5.4)	4 (1.8)	0
Cough	59 (13.4)	0	—	25 (11.2)	0	—
Pruritus	57 (12.9)	0	—	13 (5.8)	0	—
Dizziness	55 (12.5)	3 (0.7)	—	13 (5.8)	0	—
AST increased	54 (12.2)	10 (2.3)	0	15 (6.7)	6 (2.7)	0
Blood creatinine level increased	52 (11.8)	4 (0.9)	0	1 (0.4)	0	0
Arthralgia	51 (11.6)	1 (0.2)	—	32 (14.3)	1 (0.4)	—
Edema peripheral	51 (11.6)	0	—	15 (6.7)	0	—
Rash	49 (11.1)	5 (1.1)	0	10 (4.5)	0	0
URTI	49 (11.1)	0	0	17 (7.6)	2 (0.9)	0
Dyspnea	48 (10.9)	11 (2.5)	1 (0.2)	25 (11.2)	3 (1.3)	0
Pyrexia	48 (10.9)	2 (0.5)	1 (0.2)	13 (5.8)	1 (0.4)	0
Muscular weakness	47 (10.7)	4 (0.9)	—	13 (5.8)	0	—
Hot flush	46 (10.4)	0	—	22 (9.9)	0	—
Weight decreased	46 (10.4)	1 (0.2)	—	5 (2.2)	1 (0.4)	—
Back pain	42 (9.5)	3 (0.7)	—	28 (12.6)	2 (0.9)	—

Abbreviations: —, grade does not exist for this adverse event; CTCAE, Common Terminology Criteria for Adverse Events; URTI, upper respiratory tract infection.

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regarding granulocyte colony-stimulating factor administration and dose reduction was not followed, and one was due to viral pneumonia in a patient receiving steroids for spinal stenosis.

DISCUSSION

The MONARCH 2 study demonstrated that abemaciclib, a potent CDK 4 and 6 inhibitor dosed on a continuous schedule, significantly extended PFS when added to fulvestrant in women with HR+/ HER2− ABC whose disease had progressed while they were receiving ET. This benefit was consistent across subgroups.

MONARCH 2 evaluated patients whose disease progressed while they were receiving neoadjuvant or adjuvant ET, ≤ 12 months after adjuvant ET, or while receiving ET for ABC. Only 8.8% of patients progressed within 12 months after completing adjuvant therapy. Patients could not have received chemotherapy or more than one line of ET for mBC, making this a more homogeneous population than in previous studies of patients whose disease progressed while they were receiving prior ET for ABC.[5,22,24-26] The MONARCH 2 population is readily identifiable in the clinical setting. To the best of our knowledge, the median PFS of 16.4 months and the 7.2-month improvement over control therapy observed in patients who received abemaciclib plus fulvestrant represents the longest reported in a population with ABC whose disease had progressed while they were receiving prior ET.[5,22,24-26]

It is not typical for ET-based treatments to induce substantial tumor shrinkage, especially in patients whose disease has progressed while they were receiving prior ET.[27] The addition of abemaciclib to fulvestrant significantly improved ORR, which included 14 patients in the abemaciclib arm experiencing a CR. Tumor size reduction was more pronounced in the abemaciclib arm, and tumor response was durable (Fig 4). To the best of our knowledge, the ORR achieved in patients who received abemaciclib plus fulvestrant is the highest observed in a phase III study of patients whose disease had progressed while they were receiving prior ET.[5,22,24,26] Fewer patients in the abemaciclib arm experienced disease progression at the first imaging assessment, despite disease in most of the study population having progressed while they were still receiving ET.

The safety profile of abemaciclib with fulvestrant was broadly consistent with that reported for other CDK 4 and 6 inhibitors,[5,6] with the exception of diarrhea. Most diarrhea was of low grade, occurred early in the first treatment cycle, and was managed with dose adjustment and standard antidiarrheal medication. Other frequent, nonhematological AEs in the abemaciclib arm included nausea, fatigue, and abdominal pain. Severe neutropenia has been frequently reported in phase III studies with other CDK 4 and 6 inhibitors and was a dose-limiting toxicity leading to the definition of intermittent dosing schedules.[5,6,28,29] This has not been the case in MONARCH 2 and other studies of abemaciclib, in which severe neutropenia has been infrequent.[4,10,30] This may be due to the greater relative potency of abemaciclib for cyclin D1/CDK4 compared with cyclin D3/CDK6 observed in cell-free enzymatic assays.[16]

In conclusion, MONARCH 2, an evaluation for abemaciclib given at a dosage of 150 mg twice daily on a continuous schedule in combination with fulvestrant, significantly improved PFS and ORR compared with placebo plus fulvestrant and showed a tolerable safety profile in patients with HR+/HER2− ABC whose disease had progressed while they were receiving prior ET. Abemaciclib plus fulvestrant was an effective treatment for patients with HR+/HER2− ABC whose disease progressed while they were receiving ET.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at jco.org.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

MONARCH 2: Abemaciclib in Combination With Fulvestrant in Women With HR+/HER2− Advanced Breast Cancer Who Had Progressed While Receiving Endocrine Therapy

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Appendix

Table A1. Granulocyte Colony-Stimulating Factor Use

G-CSF Use*, Patients and Type	Abemaciclib + Fulvestrant (n = 441)	Placebo + Fulvestrant (n = 223)
Subjects with ≥ 1 G-CSF/GM-CSF	31 (7.0)	2 (0.9)
Filgrastim	27 (6.1)	1 (0.4)
Pegfilgrastim	1 (0.2)	1 (0.4)
Other	4 (0.9)	0

Note. Data given as No. (%)
Abbreviations: G-CSF, granulocyte colony-stimulating factor; GM-CSF, granulocyte-macrophage colony-stimulating factor
*Physicians were advised to administer G-CSF according to ASCO guidelines.

Table A2. Central Laboratory-Based Abnormalities Experienced by ≥ 10% (safety population)

	CTCAE Grade							
	Abemaciclib + Fulvestrant (n = 441)				Placebo + Fulvestrant (n = 223)			
CTCAE Term	Any	2	3	4	Any	2	3	4
Patients with ≥ 1 CTCAE term	434 (100)	178 (41.0)	187 (43.1)	28 (6.5)	216 (98.6)	62 (28.3)	23 (10.5)	4 (1.8)
Creatinine level increased*,†	427 (98.4)	191 (44.0)	5 (1.2)	0	161 (73.5)	7 (3.2)	0	0
White blood cell count decreased	384 (90.1)	182 (42.7)	96 (22.5)	3 (0.7)	71 (32.6)	23 (10.6)	2 (0.9)	0
Neutrophil count decreased	371 (87.1)	161 (37.8)	122 (28.6)	15 (3.5)	66 (30.3)	16 (7.3)	8 (3.7)	1 (0.5)
Anemia	357 (83.8)	206 (48.4)	11 (2.6)	0	73 (33.5)	17 (7.8)	1 (0.5)	0
Lymphocyte count decreased	268 (62.9)	106 (24.9)	51 (12.0)	1 (0.2)	69 (31.7)	25 (11.5)	4 (1.8)	0
Platelet count decreased	226 (53.2)	29 (6.8)	4 (0.9)	5 (1.2)	32 (14.7)	2 (0.9)	0	0
ALT level increased	178 (41.0)	35 (8.1)	17 (3.9)	3 (0.7)	71 (32.4)	9 (4.1)	3 (1.4)	0
AST level increased	162 (37.4)	15 (3.5)	17 (3.9)	0	55 (25.1)	5 (2.3)	8 (3.7)	1 (0.5)
Hyponatremia	153 (35.3)	0	17 (3.9)	0	62 (28.3)	0	6 (2.7)	0
Hypokalemia	146 (33.6)	0	30 (6.9)	1 (0.2)	25 (11.4)	0	1 (0.5)	0
Hypercalcemia	132 (30.4)	3 (0.7)	1 (0.2)	1 (0.2)	70 (32.0)	0	1 (0.5)	1 (0.5)
Hypocalcemia	105 (24.2)	19 (4.4)	2 (0.5)	3 (0.7)	43 (19.6)	7 (3.2)	0	0
ALP level increased	78 (18.0)	21 (4.8)	4 (0.9)	0	38 (17.4)	10 (4.6)	3 (1.4)	0
Hypoalbuminemia	60 (13.8)	14 (3.2)	0	0	20 (9.1)	10 (4.6)	0	0
Hyperkalemia	31 (7.1)	5 (1.2)	3 (0.7)	1 (0.2)	26 (11.9)	5 (2.3)	1 (0.5)	0

Note. Data given as No. (%).
Abbreviations: ALP, alkaline phosphatase; CTCAE, Common Terminology Criteria for Adverse Events.
*CTCAE version 4.0 defines grade 1 creatinine increase as > 1 to 1.5 times baseline or higher than the upper limit of normal to 1.5 times the upper limit of normal.
†Abemaciclib has been shown to increase serum creatinine level due to inhibition of renal tubular secretion of creatinine (a competitive inhibitor of organic cation transporter 2, multidrug and toxin extrusion protein 1, and multidrug and toxin extrusion protein 2) without affecting glomerular function.

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